

DATABASE SCHEMA DOCUMENTATION

Food Delivery Operations Analytics Pipeline

1. Schema Overview

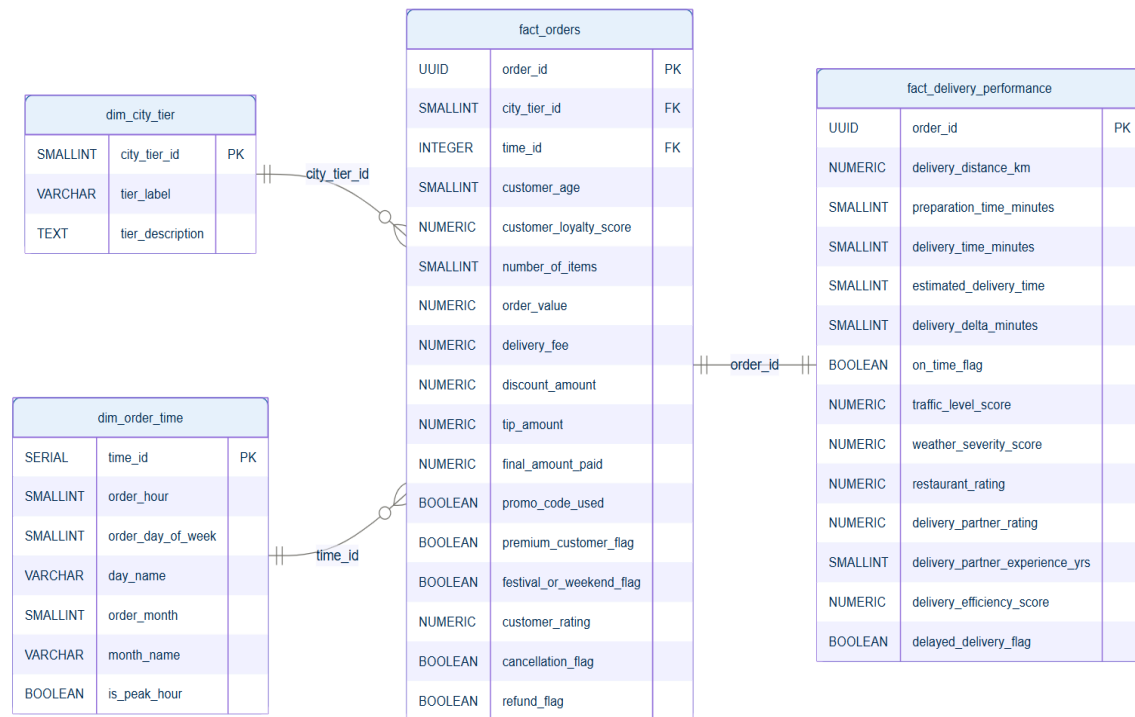
The Food Delivery Analytics database implements a star schema with two dimension tables and two fact tables. This design follows the professor's guidance from Week 2 slides: fact tables store measurable events at the order level while dimension tables provide descriptive context for slicing and filtering. The schema is optimized for the analytical queries that will power the Week 4 Dash dashboard.

Database name: food_delivery | RDBMS: PostgreSQL | Source: food_delivery_analytics_cleaned.csv (15,000 rows, 30 columns)

Table Name	Type	Row Count	Columns	Primary Purpose
dim_city_tier	Dimension	3	3	City tier reference lookup (Metro / Urban / Suburban)
dim_order_time	Dimension	~144	7	Time attributes: hour, day, month, peak-hour flag
fact_orders	Fact	15,000	17	Order financials, customer attributes, operational flags
fact_delivery_performance	Fact	15,000	14	Delivery logistics, partner metrics, engineered time deltas

2. Entity Relationship Diagram

The diagram below shows all four tables, their columns with data types, primary keys (PK), foreign keys (FK), and the crow's foot notation relationships between them. Dimension tables appear on the left and top; fact tables are connected via foreign key constraints.



Design

Star Schema Choice

A star schema was selected over a fully normalized (3NF) schema because the primary use case is analytical reporting and dashboard visualization, not transactional updates. Per the Week 2 slides, star schemas make dashboard queries faster and simpler by reducing the number of joins required at query time.

Table Decomposition

- **fact_orders**: Holds all customer-facing and financial attributes per order. Primary fact table and join hub for most dashboard queries.
- **fact_delivery_performance**: Separated from fact_orders because it captures operational/logistics attributes that analysts query independently. One-to-one relationship with fact_orders via order_id.
- **dim_city_tier**: Only 3 rows. Avoids repeating 'Tier 1 - Metro' descriptions across 15,000 fact rows, reduces redundancy, and enables easy label updates.
- **dim_order_time**: Centralizes time attributes and the engineered is_peak_hour flag. Enables time-based filtering without storing redundant hour/day/month columns in every fact row.

Engineered Columns

- **delivery_delta_minutes**: Computed as delivery_time_minutes minus estimated_delivery_time. Positive values indicate late deliveries; negative values indicate early. Dataset on-time rate: 90.53%.

- **on_time_flag:** Boolean derived from `delivery_delta_minutes <= 0`. Pre-computing this avoids repeated CASE WHEN logic in dashboard queries.
- **is_peak_hour:** TRUE for order hours 11-13 (lunch) and 18-21 (dinner). Enables peak vs. off-peak segmentation in a single filter.

4. Table Definitions

4.1 dim_city_tier

Type: Dimension | Rows: 3 | Purpose: City tier reference lookup

Column Name	Data Type	Nullable	PK	FK	Description
city_tier_id	SMALLINT	NO	YES		Primary key. Values: 1 (Metro), 2 (Urban), 3 (Suburban)
tier_label	VARCHAR(50)	NO			Human-readable label e.g. 'Tier 1 - Metro'
tier_description	TEXT	YES			Long-form description of city tier characteristics

4.2 dim_order_time

Type: Dimension | Rows: ~144 | Purpose: Time attributes with peak-hour engineering

Column Name	Data Type	Nullable	PK	FK	Description
time_id	SERIAL	NO	YES		Surrogate primary key; auto-generated
order_hour	SMALLINT	NO			Hour of day (0-23)
order_day_of_week	SMALLINT	NO			Day of week (1=Monday through 6=Saturday)
day_name	VARCHAR(15)	YES			Derived text label e.g. 'Monday', 'Tuesday'
order_month	SMALLINT	NO			Calendar month (1-12)
month_name	VARCHAR(15)	YES			Derived text label e.g. 'January', 'June'
is_peak_hour	BOOLEAN	NO			Engineered flag: TRUE if hour is 11-13 or 18-21 (lunch/dinner peak)

4.3 fact_orders

Type: Fact | Rows: 15,000 | Purpose: Order financials, customer attributes, and operational flags

Column Name	Data Type	Nullable	PK	FK	Description
order_id	UUID	NO	YES		Primary key; UUID from source dataset

Column Name	Data Type	Nullable	PK	FK	Description
city_tier_id	SMALLINT	NO		YES	FK -> dim_city_tier.city_tier_id
time_id	INTEGER	NO		YES	FK -> dim_order_time.time_id
customer_age	SMALLINT	YES			Customer age in years (range: 18-75)
customer_loyalty_score	NUMERIC(8,6)	YES			Continuous loyalty score (0.0-10.0)
number_of_items	SMALLINT	YES			Items per order (range: 1-12)
order_value	NUMERIC(10,4)	NO			Pre-discount order value in USD (\$100-\$215.61)
delivery_fee	NUMERIC(8,4)	NO			Delivery fee charged in USD
discount_amount	NUMERIC(8,4)	NO			Discount applied to order in USD
tip_amount	NUMERIC(8,4)	YES			Customer tip in USD; 150 records are NULL
final_amount_paid	NUMERIC(10,4)	NO			Net amount paid after fees, discounts, tips
promo_code_used	BOOLEAN	NO			TRUE if promo code applied; 42.31% of orders
premium_customer_flag	BOOLEAN	NO			TRUE if customer holds premium membership
festival_or_weekend_flag	BOOLEAN	NO			TRUE if order placed on festival day or weekend
customer_rating	NUMERIC(3,1)	YES			Customer satisfaction rating 1.0-5.0; 150 NULLs
cancellation_flag	BOOLEAN	NO			TRUE if order was cancelled; 13.35% of orders
refund_flag	BOOLEAN	NO			TRUE if refund was issued; 4.12% of orders

4.4 fact_delivery_performance

Type: Fact | Rows: 15,000 | Purpose: Delivery logistics, partner performance, and engineered delta metrics

Column Name	Data Type	Nullable	PK	FK	Description
order_id	UUID	NO	YES	YES	PK + FK -> fact_orders.order_id (one-to-one)
delivery_distance_km	NUMERIC(8,4)	NO			Distance from restaurant to customer in km
preparation_time_minutes	SMALLINT	NO			Restaurant food prep time in minutes
delivery_time_minutes	SMALLINT	NO			Actual end-to-end delivery duration in minutes
estimated_delivery_time	SMALLINT	NO			System-estimated delivery time at order placement

Column Name	Data Type	Nullable	PK	FK	Description
delivery_delta_minutes	SMALLINT	NO			ENGINEERED: actual minus estimated delivery time
on_time_flag	BOOLEAN	NO			ENGINEERED: TRUE if delivery_delta_minutes <= 0
traffic_level_score	NUMERIC(4,1)	NO			Traffic congestion score at time of order (0-10)
weather_severity_score	NUMERIC(4,1)	NO			Weather impact score at time of order (0-10)
restaurant_rating	NUMERIC(3,1)	YES			Restaurant quality rating 1.0-5.0
delivery_partner_rating	NUMERIC(3,1)	YES			Partner performance rating 1.0-5.0; 150 NULLs
delivery_partner_experience_years	SMALLINT	NO			Years of experience (range: 1-15)
delivery_efficiency_score	NUMERIC(6,2)	NO			Composite efficiency score 1.9-100.0 (higher = better)
delayed_delivery_flag	BOOLEAN	NO			TRUE if order marked as delayed; 9.47% of orders

Table Relationships

- **fact_orders.city_tier_id -> dim_city_tier.city_tier_id:** Many-to-one. Each order belongs to one city tier.
- **fact_orders.time_id -> dim_order_time.time_id:** Many-to-one. Each order maps to a unique time dimension record.
- **fact_delivery_performance.order_id -> fact_orders.order_id:** One-to-one. Every order has exactly one delivery performance record.

Null Value Audit

Three columns in the source CSV contain 150 null values each (1.0% of records). These are retained as NULL in PostgreSQL to preserve data fidelity. Imputation will occur in the transformation layer (Week 3) for specific dashboard queries where nulls would distort aggregations.

Column	Null Count	% of Records	Handling Strategy
customer_rating	150	1.00%	Retain as NULL in PostgreSQL; impute with median (4.0) in Dash queries
delivery_partner_rating	150	1.00%	Retain as NULL; excluded from avg rating KPI calculations
tip_amount	150	1.00%	Retain as NULL; treat as \$0 tip in revenue aggregations

Indexes

Six indexes are created at load time to support dashboard queries without full table scans:

- `idx_fact_orders_city_tier`: Supports city-tier segmentation filters on `fact_orders`
- `idx_fact_orders_time`: Supports hour/month/day joins from `dim_order_time` to `fact_orders`
- `idx_fact_orders_premium`: Supports premium vs. non-premium customer segmentation
- `idx_fact_orders_cancelled`: Supports cancellation rate calculations and filtering
- `idx_perf_efficiency`: Supports delivery efficiency scatter plot queries
- `idx_perf_on_time`: Supports on-time rate KPI and heatmap filtering

Key Business Metrics Derivable from Schema

- **Cancellation Rate:** `SELECT AVG(cancellation_flag::int) FROM fact_orders` -> 13.35%
- **On-Time Delivery Rate:** `SELECT AVG(on_time_flag::int) FROM fact_delivery_performance` -> 90.53%
- **Avg Revenue per Order:** `SELECT AVG(final_amount_paid) FROM fact_orders` -> \$119.04
- **Partner Efficiency by Experience:** `SELECT delivery_partner_experience_years, AVG(delivery_efficiency_score) FROM fact_delivery_performance GROUP BY 1 ORDER BY 1`
- **Peak Hour Cancellation Rate:** `JOIN fact_orders to dim_order_time on time_id, filter is_peak_hour = TRUE, AVG(cancellation_flag)`

PGADMIN4 OUTPUT

